

Tutorial 6

CSE 112 Computer Organization

ISA description:

The instructions supported by the ISA are mentioned in the table below. The ISA has 16 general purpose registers: **r0 to r15**

Name	Semantics	Syntax
Add	Performs $\text{reg1} = \text{reg2} + \text{reg3}$	<code>add reg1, reg2, reg3</code>
Sub	Performs $\text{reg1} = \text{reg2} - \text{reg3}$	<code>sub reg1, reg2, reg3</code>
Branch if equal	Branch to addr if $\text{reg1} = \text{imm}$	<code>beq reg, #imm, addr</code>
Mov reg	Performs $\text{reg2} = \text{reg1}$	<code>mov reg2, reg1</code>
Mov imm	Moves imm, a constant value, to reg	<code>mov reg, #imm</code>

Apart from the above instructions, the assembler and the operating system support the following subroutines.

Name	Semantics	Syntax
Input	Reads immediate data from user into reg	<code>in reg</code>
Output	Prints str on the console	<code>out "str"</code>
Halt	Halts the program execution	<code>hlt</code>

Q1: Nested Loops

Write an assembly program to print the following pattern for **n** lines, where **n** is read as input from the user. Note that performing **out "\n"** moves the cursor to the next line. Assume that **n** will always be greater than or equal to 1.

```
* * * * *
* * * *
* * *
* *
*
```

(Hint: The above pattern would be printed when $n = 5$)

Q2: Switch Case Without Break

Write an assembly program that takes an input **n** from the user and prints

n n-1 n-2 n-3 ... 0

If **n** is **0**, then only **0** is printed.

Assume that **n** will always be one of **0, 1, 2, 3, 4, or 5**.

Example 1: If n=3, print

3 2 1 0

Example 2: If n = 1, print

1 0

Use switch-case like structures in your solution.

Q3: Switch Case With Break

Write an assembly program that takes an input **n** from the user and prints back the number.

Valid values of **n** are **0, 1, 2, 3, 4, or 5**. If any other value is given to **n**, print **"Invalid"**.

Example 1: If n=3, print

3

Example 2: If n = 1, print

1

Example 3: If n = 10, print

Invalid

Use switch-case like structures in your solution.
